

9. CRITICAL EVALUATION OF RESEARCH METHODOLOGIES

This section must include a critical evaluation of research methodologies and processes used in this computing research project, justification of the methods and analysis chosen. This section discusses:

- Choice and philosophical justification of methodologies;
- Application of methodologies;
- Effectiveness of methodologies in addressing each RQ;
- Rigor of analytical processes; and
- Alternative designs and what they would have yielded.

9.1 JUSTIFICATION OF RESEARCH PHILOSOPHY

The positivist philosophy was selected because all five research questions are phenomena that are measurable and able to be tested empirically. F1 scores are objective, Likert distributions are quantifiable, drift degradation percentages are calculable. The quantitative benchmarking strand would not fit well with a constructivist or interpretivist philosophy.

The deductive approach (hypothesis generation from existing theory and testing of hypotheses against data) was used because the literature contained clear directional predictions (AI should outperform signature IDS; governance barriers should hinder FL) that could be tested against data. This is in line with the scientific method (Stallings, 2023)

9.2 CRITICAL EVALUATION OF MIXED-METHODS DESIGN

A concurrent mixed methods design was selected over a quantitative or qualitative design for three important reasons. First, completeness: No one method can answer all 5 RQs; RQ1 and RQ4-5 require simulation; RQ2-3 require practitioner survey. Second, triangulation: concurrent design enables cross strand validation. The most significant is the adversarial concern divergence with Q13 mean 2.82/5 (survey) and PGD 63% F1 collapse (simulation). This difference, which can be detected only by simultaneous techniques, is a significant discovery in itself. Third, richness: Q15 open text provides qualitative validation which cannot be achieved with Likert data.

One limitation of the concurrent design is that the survey was created prior to the simulation results and therefore concept drift questions were not incorporated.

9.3 CRITICAL EVALUATION OF SURVEY INSTRUMENT

Strengths: Six Likert items on consistent 1–5 scales enable direct cross dimensional comparison. Multi-select Q7 offers more detailed barrier profiling. Independent qualitative validation is provided with Q15 open text.

Weaknesses:

1. Q10 is in a categorical format and cannot be used to compute the mean.

2. No concept drift question (coverage gap found only after simulation).
3. Question order can prime respondents (Q6 before Q7).
4. No reverse-scored Likert items (acquiescence bias).

Analytical Approach: Means, medians, modes, SDs, and full score distributions for each Likert dimension is suitable for the descriptive study with n=22. When reporting full distributions, rather than means alone, this avoids the possibility of reporting bimodal Q6 data as moderate ambivalence.

9.4 CRITICAL EVALUATION OF ALGORITHM BENCHMARKING

The most methodologically rigorous strand of this study is the benchmarking methodology: OT-specific datasets (SWaT), SMOTE, five-fold cross-validation. The SWaT dataset use is a true methodological improvement compared to enterprise IT datasets found in the published literature.

SMOTE Application: Very important and sometimes overlooked. If SMOTE is not used, supervised algorithms will yield artificially high accuracy levels by predicting the majority class. Correct metric used for imbalanced classes: F1 score (equally weighting precision and recall) and methodologically justified. The results of SMOTE were 0.12-0.19 improvement over F1 for supervised algorithms, which validates the use of SMOTE. Latency Benchmarking: Latency added with F1 adds practical deployment value not found in most academic benchmarking. The 100-200ms OT threshold is a real constraint; One-Class SVM at 95ms is getting close to this limit, irrespective of its accuracy. This deployment-based assessment is a form of critical thinking that goes beyond the standard benchmarking assessment.

Concept Drift Methodology: The methodology of Concept Drift is original and has no direct literature precedent, the drift simulation was performed for 30 weeks, with 10 discrete events. The quantification of the 29.4% F1 silent degradation gives new, practically important risk parameters.

9.5 OVERALL METHODOLOGICAL RIGOUR ASSESSMENT

Research Question	Methods Applied	Rigour Assessment	Confidence	Key Limitation
RQ1: AI vs Signature IDS on novel OT threats	Algorithm benchmarking (SWaT CIC-IDS; SMOTE; 5-fold CV; 8 algorithms)	High OT datasets; latency tested; novel threat F1 isolated	Very High3/3 strands converge	Digital twin gap; ~10–15% performance discount needed

Research Question	Methods Applied	Rigour Assessment	Confidence	Key Limitation
RQ2: XAI impact on operator trust	Survey Q9/Q10 + literature (Schneiderman, 2020)	Moderate-High6-point Likert; Q10 categorical limit	High survey corroborates literature; Mode=5 strong signal	No experimental trust measurement; early-career skew
RQ3: FL adoption barriers	Survey Q11/Q12/Q14 + literature (Kizza, 2023)	High54.5pp gap quantified; all dimensions measured	High Kizza (2023) directly corroborated	No FL pilot; governance prerequisites not operationally tested
RQ4: Concept drift impact	30-week simulation + Q15 open text validation	High novel; 10 events; drift rates quantified	Medium-High simulation only; Q15 partial cross-validation	Discrete events only; continuous drift not modelled
RQ5: Adversarial attack risk	10-type adversarial testing + survey Q13	High500 samples; composite risk ranking	High critical knowledge-risk gap confirmed	Compound attacks not tested; live adversary intelligence absent

The concurrent mixed-methods design is well justified and suitable for all five research questions and results in original contributions that are not available in the literature. The algorithm simulation strand is the most methodologically rigorous and results in new OT-specific benchmarks. The survey strand has sample size and experience skew limits which are disclosed and mitigated in each section. Given the limitations of each strand, the concurrent triangulation of all three strands significantly enhances confidence in the five major findings.